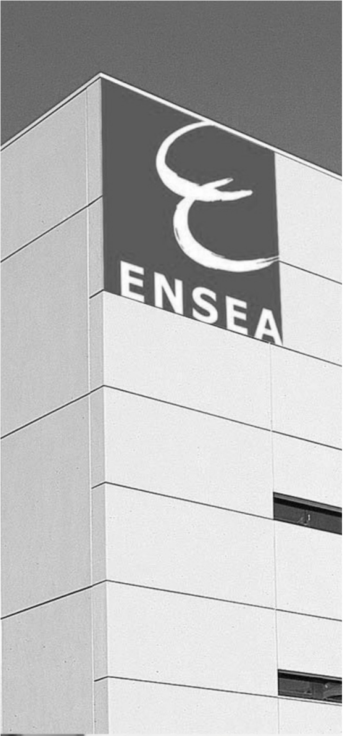
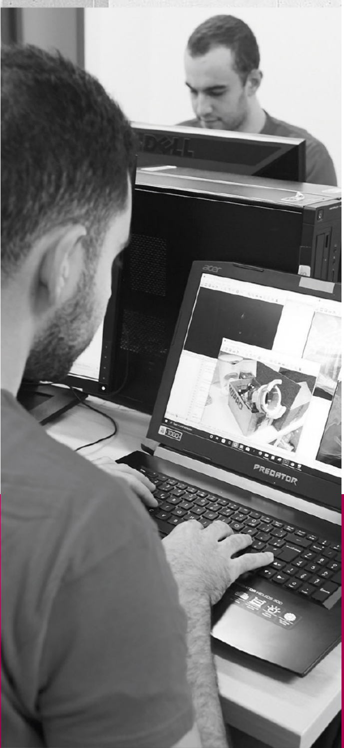





ENSEA

Beyond Engineering

POWER ELECTRONICS

DEE_2401

TP

Semestre 7

Alexis MARTIN
2026

1 Recommendations for the Labs

The purpose of these labs is to illustrate the second year course on energy conversion through the study of four experiments:

- Buck converter (simulations with LTSpice)
- Four-quadrant chopper
- Flyback switching power supply
- Forward switching power supply

In order to prepare for these labs, it is necessary to know the corresponding course and to have read the complete lab text in order to plan the different steps of the study. In particular, when a comparison with theoretical curves is requested, it is necessary to bring them to the lab and to know how to adapt them to the lab.

The lab topics are conducted in rotation: it is therefore imperative to know your binomial number before coming to the lab.

	Pair 1	**Pair 2**	**Pair 3**	**Pair 4**	**Pair 5**	**Pair 6**	**Pair 7**
Session 1	4Q Chop.	4Q Chop.	4Q Chop.	LTSpice	LTSpice	LTSpice	LTSpice
Session 2	LTSpice	LTSpice	LTSpice	4Q Chop.	4Q Chop.	4Q Chop.	4Q Chop.
Séance 3	Forward	Forward	Forward	Flyback	Flyback	Flyback	Flyback
Séance 4	Flyback	Flyback	Flyback	Forward	Forward	Forward	Forward

Depending on the laboratory where the lab sessions take place, some differences may appear in the wiring of the machines during the four-quadrant chopper lab as well as in the nominal values of these machines which are shown on their nameplate and which must therefore be carefully noted.

<u>At the end of each session</u>, a report must be submitted to the teacher. The evaluation will take into account both the quality of the report and the behavior during the session itself.

2 Buck Converter

The lab follows the tutorial on the Buck converter, whose topic is recalled below. The goal is to use the LTSpice software to simulate the circuit and understand its operation. LTSpice is a software provided by Analog Device, based on Spice simulation. The software documentation is available on the Analog Device website: Analog Device Website

The goal is to power a STM32H4 microprocessor using a battery. The specifications are as follows:

- LiPo 2S battery $V_{in} = 7.4\ V$
- Microprocessor supply voltage: $V_{out} = 3.3\ V$
- Maximum voltage ripple at the microprocessor input: $\Delta V_S = 50\ mV$
- Ratted output current: $I_S = 200\ mA$
- Current ripple: 30%

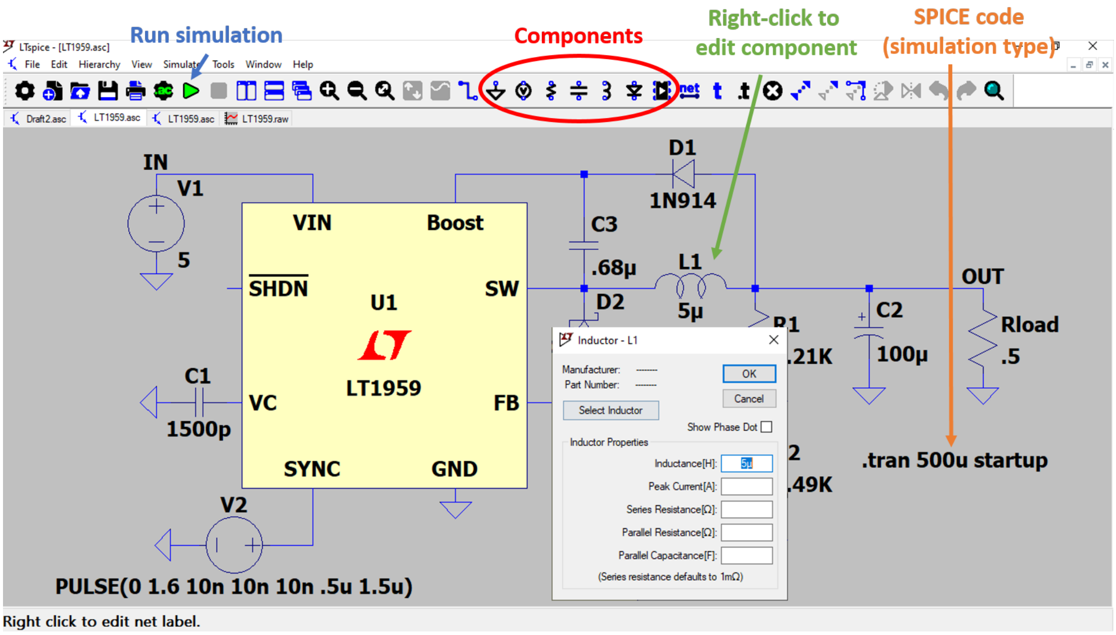
For this, we want to use the following component: **LT1959**, an excerpt from the documentation is in the appendix. In this exercise, we will assume that the **components are ideal**, and that the Buck operates in **continuous conduction mode**.

2.1 Preparation

1. Draw the circuit to use this component in Buck operation
2. Recall the values of (no calculation is required):
   - the switching frequency of the chopper
   - the values of L_1, C_1 of the output filtering of the chopper to match the specifications
   - the resistances R_1 and R_2 of feedback to have an output voltage of $3.3\ V$
   - the value of the resistance R_m modeling the microcontroller absorbing a rated current
   - the values of R and C for the output voltage control limiting the duty cycle at startup to $\frac{\alpha_n}{2}$ and a time constant of the corrector 10 times greater than the time constant of the uncorrected system

2.2 Circuit Simulation

Here is a brief summary of the LTSpice interface. For more information, you can refer to the Analog Device page: Analog Device Website



- Create a new *Schematic* and place the **LT1959**.
- Create the circuit from the datasheet by right-clicking on the component ⇒ "*Open Example Circuit*"
- Modify the circuit to match the datasheet. The "*SYNC*" pin will not be used.
- Modify the component values to match the specifications.
- Perform a transient simulation. To do this, go to "*Simulate*" ⇒ "*Configure Analysis*" ⇒ "*Transient*" and perform a simulation over 3 ms.

1. Check the value of the switching frequency of the converter
2. Check that the specifications are met: output voltage, current ripple, voltage ripple
3. Is the converter operating in continuous or discontinuous conduction mode?

2.3 Influence of Components

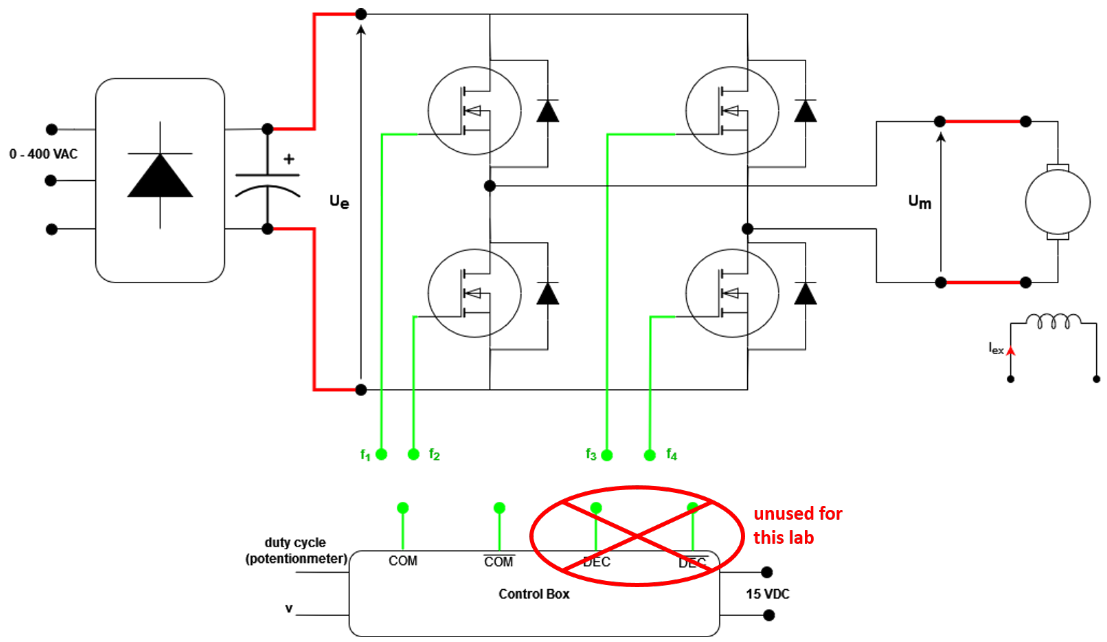
1. Measure the value of the current ripple in L_1 and the output voltage ripple for different values of L_1. Comment.
2. Measure the value of the output voltage ripple for different values of C_1. Comment.
3. Determine the value of L_1 to be in critical conduction mode. Verify by simulation.

2.4 System Speed and Stability

1. Modify the value of C and measure the 5% time response. Comment.
2. By modifying the value of R, determine the maximum value of the resistance from which the system becomes unstable.
3. Add a current generator in parallel with the resistance R_s to generate a current pulse (once the system is in steady state). This pulse can model an EMC disturbance on the system. Determine the new value of R from which the system becomes unstable during a disturbance.

3 4 quadrants chopper

The general scheme of the device is shown in the following figure:



The supply U_E is provided by a three-phase rectification of the 50 Hz network, variable up to 400 V using an autotransformer, followed by a filtering capacitor of value C. The DC machine has independent excitation, which will be kept constant with an excitation current equal to its rated value.

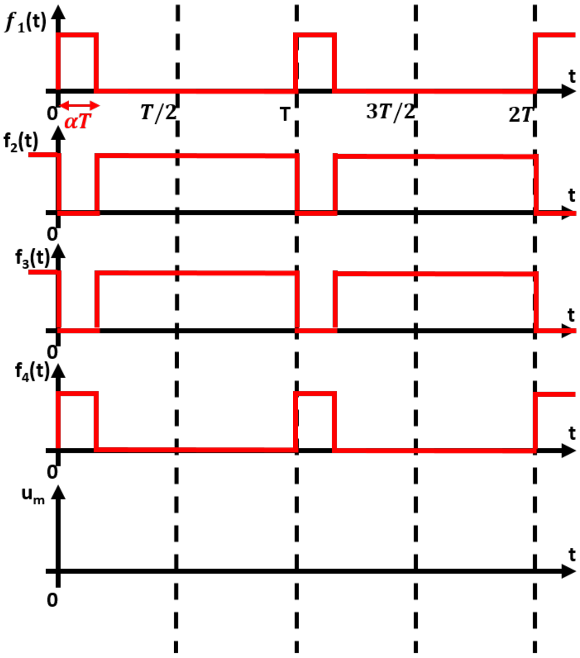
The "rectifier-chopper" assembly is implemented within an integrated SEMIKRON device equipped with an external control system that allows for complementary synchronous or shifted control on each of the two "bridge arms".

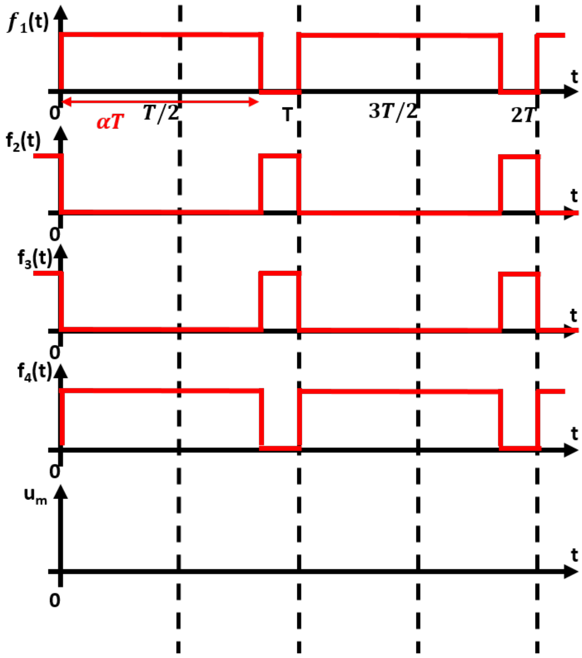
The control device generates PWM signals whose duty cycle can be controlled using a potentiometer or an external voltage $v(t)$ **between 0 and 5V**.

3.1 Preparation

The chopper is controlled with the control signals f_1, f_2, f_3 and f_4 below.

1. Plot the evolution of the chopper output voltage $U_m(t)$ assuming that the input voltage $U_e(t)$ is perfectly constant.
2. Calculate the average value of $U_m(t)$ as a function of the input voltage U_e (assumed constant) and α.





3.2 Assembly of the circuit

1. Read the characteristics of the DC machine and deduce the value of the rectified DC voltage U_E necessary to exploit the entire speed range of this machine in both directions. This "theoretical" value of U_E will be increased by 20% to take into account voltage drops and duty cycle limitations inherent to the control box.

2. Observe the shape of the control signals labeled COM and $\overline{COM}$ on the control box (the signals DEC and $\overline{DEC}$ will not be used). **Ensure that the duty cycle is different from 50%**. Deduce:
   - The connection of the control signals COM and $\overline{COM}$
   - The shape of the chopper output voltage, particularly its amplitude and frequency.

3. Plot the control characteristic of this command, i.e. the duty cycle of the COM command as a function of the external control voltage (**variable in the range 0 to 5V**). Deduce the value of this voltage that imposes the stop of the machine.

4. Assemble the complete circuit with the following recommendations:
   - Leave the DC machine unloaded;
   - If the machine has separate excitation, first wire its excitation and check the corresponding current value;
   - Adjust the chopper control to impose a duty cycle of 50%;
   - Gradually increase the network voltage (autotransformer) until the desired rectified voltage U_E is reached;
   - Test the proper functioning of the device by varying the chopper control **slowly** in both directions while continuously monitoring the motor armature current.

   The assembly being quite "bulky" to implement, care should be taken to wire the different blocks together in a way that is easily verifiable. The various measuring devices should be readable, and the oscilloscope should be as far away as possible from the power connections.

5. If the motor braking is too fast, the DC bus voltage U_E increases. Why?
   <u>In the presence of the instructor</u>, observe this phenomenon.

3.3 Static Characteristics of the Device

Always ensuring that U_E remains constant, plot the chopper characteristic $< U_m >= f(\alpha)$, with the motor left unloaded (α represents the duty cycle of the COM command). Carefully record the extreme values of possible duty cycles, $\alpha_{\min}$ and $\alpha_{\max}$.

3.4 Study of the Chopper in Dynamic Regime

In this study, a duty cycle of the type $\alpha(t) = \alpha_0 + \Delta\alpha(t)$ must be generated, where $\Delta\alpha(t)$ is a "slow" variation and $\alpha_0 = 0.5$. To achieve this, the control voltage used to drive the device includes a DC component v_0 and a low-frequency AC component, denoted $\Delta v(t)$. The motor will always be left unloaded **and care will be taken not to switch the functions of the control generator with a powered circuit** to avoid overcurrents.

1. Use the previous control characteristic to determine v_0 and $\Delta v(t)$ in order to linearly vary the duty cycle between 0.3 and 0.7. Apply this voltage gradually to the device (slowly increasing amplitude), at a cycle frequency of 0.3 Hz (frequency of the Δv signal).

2. Simultaneously record the motor armature current i_M and the voltage measured by the tachogenerator v_T (image of the speed Ω). How to interpret these curves using the XY mode of the oscilloscope (with sufficient persistence)? The measurement can be improved by inserting a low-pass filter on the current measurement to eliminate its ripple. How to choose the cutoff frequency of this filter?

3.5 To go further: Inverter

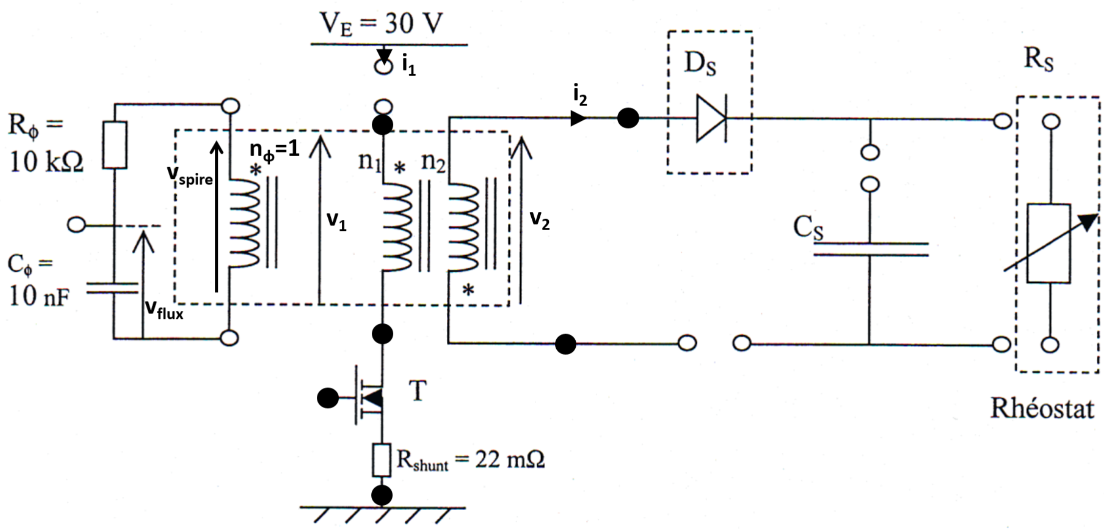
Replace the motor with a series R-L load. We want to power this load with the North American power grid (110V / 60 Hz) from our power grid (230V / 50 Hz). We will therefore use the chopper as a single-phase "inverter" to recreate this network. For this, we will use a duty cycle of the form $\alpha(t) = \alpha_0 + \delta\alpha.\sin(\omega t)$.

1. Determine α_0, $\delta\alpha$ and ω in order to recreate the desired power grid.

2. Apply this command and power the load. Observe the voltage U_m at the chopper output and U_r the voltage across the resistor.

4 Flyback Power Supply

The hardware setup includes, in addition to the transistor control circuit, the following assembly:



Test points • allow visualization of the different signals, including the control signal of the MOSFET transistor.

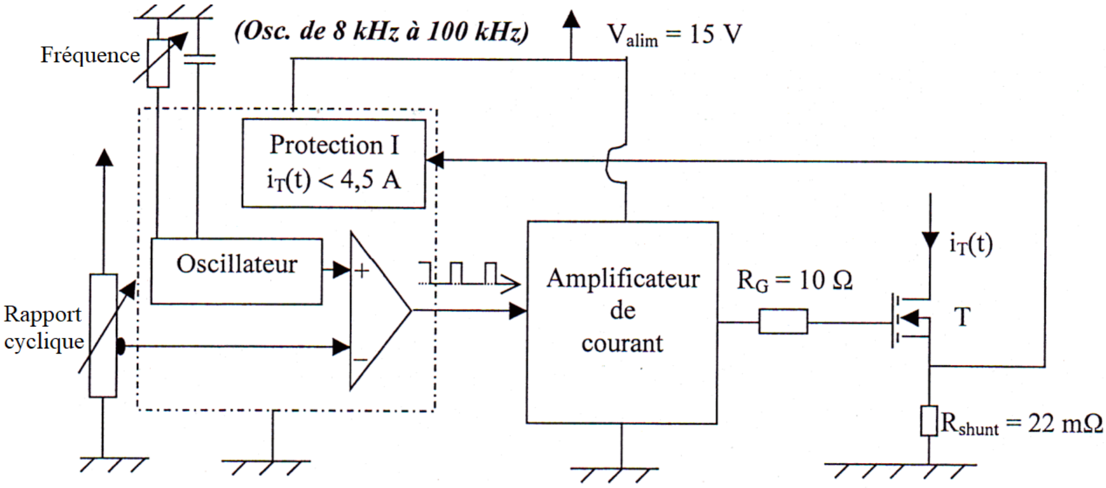
The 3 windings are on the same ETD29/16/10 magnetic core made of N27 type ferrite, which documentation is in the appendix. This core can be equipped with an air gap adjusted by non-magnetic shims. In this case, the relationship between the thickness s of the air gap and the value A_L of the specific inductance is given by the following relation where K_1 and K_2 are two constants given in the documentation:

$$s = \left(\frac{A_L}{K_1}\right)^{\dfrac{1}{K_2}}$$

The missing connections must be made by "jumpers" allowing the currents to be visualized by the use of appropriate probes. The load rheostat can be either 250 Ω or 330 Ω and care should be taken **not to connect the voltage V_E without the presence of this load**.

It should be noted that there is a flux measurement circuit made up of a single turn connected to an R-C integrator circuit. The voltage v_{flux} therefore allows the ripple of the flux in the magnetic circuit to be visualized.

The transistor control device is shown in the figure below:

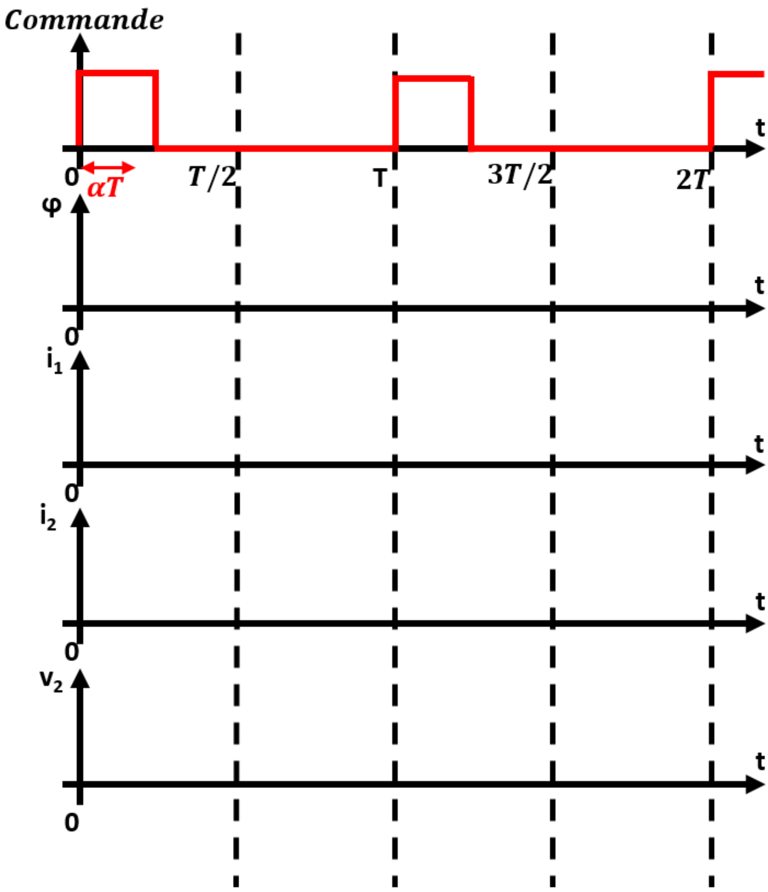


A potentiometer allows the frequency to be adjusted (do not go below 15 kHz), a second allows the duty cycle to be adjusted. The latter is equipped with an active current limit set to 4.5A and implemented by measuring the voltage across a 22 mΩ shunt. The power supply for this control is provided by an independent voltage source of value $V_{alim} = 15V$.

4.1 Preparation

We assume that the transistor is controlled by a PWM signal with a duty cycle of $\alpha = 0.25$. We also assume that the converter operates in **total demagnetization mode** for the transformer. The input voltage V_E is assumed to be constant. For preparation, we will only study the steady state.

1. Complete the chronograms below by drawing:
   - the flux φ in the transformer
   - the current i_1 delivered by the power supply
   - the current i_2 in the diode
   - the voltage v_2 at the output of the transformer



2. We assume that the time constant of the filter $R_\phi - C_\phi$ is large compared to the switching period of the chopper. Show that v_{flux} allows visualizing the flux ripple in the transformer. Determine the flux ripple $\Delta\varphi$ as a function of α, T and the value of v_{spire} over the interval $0 < t < \alpha T$.

For the entire lab session, the following precautions must be taken:

- use two independent sources to realize V_{alim} and the "power" supply V_E: **the current limit will be carefully set to 2 A**
- keep the voltage V_{alim} present during the entire session but reduce the voltage V_E at each connection modification
- do not lower the switching frequency below 15 kHz

4.2 Magnetic Study, Waveforms

Perform the following measurements with the load rheostat at its maximum value (reduced load) and by ensuring to work in the total demagnetization mode, for a duty cycle close to its maximum value.

1. Set the switching frequency to 50 kHz and the duty cycle to its maximum value (which will be noted). Observe the voltage across the flux measurement turn v_{spire} as well as the voltage v_{flux}. Deduce the peak-to-peak excursion ΔB of the magnetic induction in the circuit using the documentation ($\Delta B = B_{\max} - B_{\min}$).
2. By comparing the voltages v_{spire} and v_1, then v_2, determine the number of turns n_1 and n_2.
3. Observe simultaneously the currents i_1 and i_2 and compare them to the evolution of the flux.
4. Deduce the value of the magnetizing inductance at the primary, denoted L_1.
5. From the measurements of n_1 and L_1 and using the documentation, determine the value of the specific inductance A_L and then the thickness of the air gap. What is the role of this air gap?

4.3 Static Curves

The purpose of these measurements is to record the network of static characteristics at the output of the power supply. The switching frequency is initially kept constant at 50 kHz.

1. For several values of the duty cycle (for example $\alpha_{\max}$, 0.3, 0.2 and 0.1), modify the resistance of the load rheostat to vary the average current delivered between its minimum value (obtained with the rheostat at maximum) and a maximum value of 2 A. Then record the average values of the current delivered by the power supply V_E, I_E, the output current, I_S, and the output voltage, V_S. Carefully note the operating point corresponding to the change in regime of the power supply with respect to demagnetization.

2. Deduce the plot of the characteristics $V_S(I_S)$ and $P_S(I_S)$ (output power) parameterized by α. Also plot the power transfer characteristics $P_S(P_E)$ and deduce the efficiency. Show the operating zones in partial demagnetization and total demagnetization.

4.4 Additional Measurements

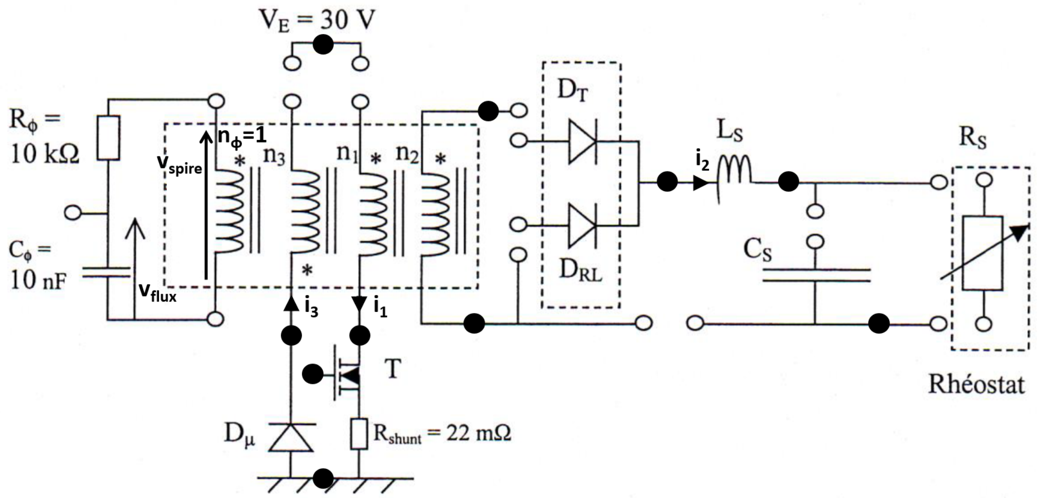
Keep the load rheostat at its maximum value and the switching frequency at 50 kHz.

1. Observe simultaneously the current i_1 and the voltage across the transistor v_T. Deduce the stresses experienced by this transistor.

2. Observe simultaneously the current in the filter capacitor I_{CS} and the output voltage v_S. Deduce the value of the filter capacitor C_S.

5 Forward Converter

The hardware setup includes, in addition to the transistor control circuit, the following setup:



Test points • allow to visualize the different signals, including the control signal of the MOSFET transistor.

The 4 windings are on the same ETD29/16/10 magnetic core made with a type N27 ferrite whose document is in the appendix. This core can be equipped with an air gap adjusted by non-magnetic shims. In this case, the relationship between the thickness s of the air gap and the value A_L of the specific inductance is given by the following relation where K_1 and K_2 are two constants given in the documentation:

$$s = \left(\frac{A_L}{K_1}\right)^{\frac{1}{K_2}}$$

The missing connections must be made by "jumpers" allowing to visualize the currents by using appropriate probes. The load rheostat can be 250 Ω or 330 Ω indifferently. Care must be taken to **never disconnect the demagnetization winding**.

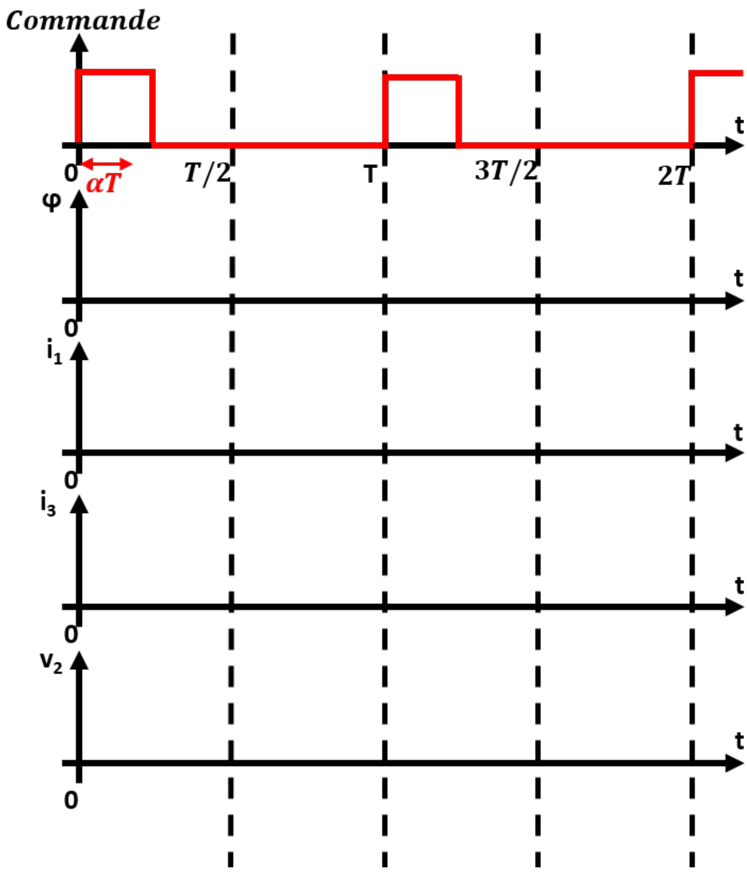
Note the presence of a flux measurement circuit made by a single turn connected to an R-C integrator circuit. The voltage v_{flux} thus allows to visualize the ripple of the flux in the magnetic circuit.

The transistor control device is identical to that of the Flyback power supply (Cf Lab "Flyback Power Supply")

5.1 Preparation

Assume that the transistor is driven by a PWM signal with a duty cycle $\alpha = 0.25$. We also assume that the converter operates in **total demagnetization mode** for the transformer and that no load is connected at the output of the converter. The input voltage V_E is assumed to be constant. We will study only the steady-state for the preparation.

1. Complete the chronograms below by plotting:
   - the flux φ in the transformer
   - the current i_1 delivered by the power supply
   - the current i_3 in the demagnetization winding
   - the voltage v_2 at the output of the transformer



2. We assume that the time constant of the filter $R_\phi - C_\phi$ is large compared to the switching period of the chopper. Show that v_{flux} allows to visualize the flux ripple in the transformer. Determine the flux ripple $\Delta\varphi$ as a function of α, T and the value of v_{spire} on the interval $0 < t < \alpha T$.

During the entire lab session, the following precautions must be taken:

- use two independent sources to realize V_{alim} and the "power" supply V_E: **carefully set the current limit of the latter to 2 A**
- keep the voltage V_{alim} present during the entire session but lower the voltage V_E at each modification of the connection
- do not lower the switching frequency below 15 kHz
- when several voltage measurements are made at the same time with a voltage probe, **be careful not to create a short circuit**. Indeed, the voltage probes are not isolated from each other, so the reference potentials of the probes are therefore linked via the oscilloscope
- **never disconnect the demagnetization winding**

5.2 No-load operation

In this first part of the study, there is no load at the output of the power supply.

1. Set the switching frequency to 50 kHz and the duty cycle to its maximum value (which we will note). Observe the voltage across the flux measurement coil v_{spire} as well as the voltage v_{flux}. Deduce the peak-to-peak excursion ΔB of the magnetic induction in the circuit with the help of the documentation ($\Delta B = B_{\max} - B_{\min}$).
2. Verify the complete demagnetization at each period by simultaneously observing the currents i_1 and i_3. Compare these observations to that of the voltage v_{flux}.
3. By simultaneously observing the current i_1 and the voltage v_1, determine the value of the magnetizing inductance at the primary, denoted L_1.
4. By comparing the voltages v_{spire} and v_1, then v_2 and v_3 (the voltages v_i are measured across the windings of n_i turns) determine the number of turns n_1, n_2 and n_3.
5. Deduce the value of the air gap used s.
6. By simultaneously observing the current i_1 and the voltage v_{flux}, gradually decrease the switching frequency down to 20 kHz while maintaining the maximum duty cycle. What happens?

5.3 Loaded operation

The load rheostat is connected after setting it to a value of 30Ω to perform the following measurements. Do not forget to connect all the "open" connections allowing the measurements of the different currents at the secondary. The switching frequency is set again to 50 kHz and the duty cycle to its maximum value.

1. Observe simultaneously and then successively the following pairs of quantities:
   - the secondary voltage v_2 and the voltage across D_{RL}, v_{DRL}
   - the output current i_S and the voltage across D_{RL}
   - the currents in the two secondary diodes, i_{DRL} and i_{DT}

   What can be said about the operating mode of these secondary diodes?

2. Observe simultaneously the current i_{LS} in the coil L_S and the voltage v_{DRL}. Deduce the value of this output inductance L_S. What can be said about its sizing?

3. Observe simultaneously the current in the filter capacitor i_{CS} and the output voltage v_S. Deduce the value of the filter capacitor C_S.

5.4 Static curves

The purpose of these measurements is to record the network of static characteristics at the output of the power supply. The switching frequency is kept constant at 50 kHz.

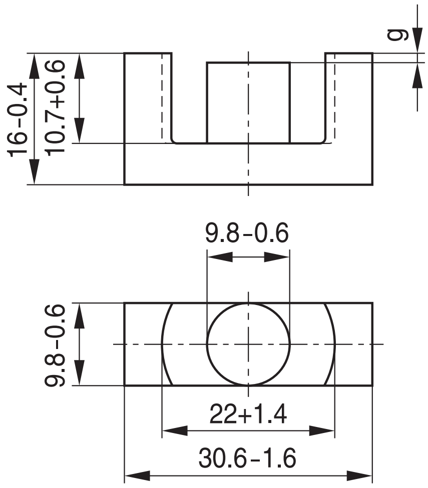
1. For several values of the duty cycle (for example $\alpha_{\max}$, 0.3, 0.2, 0.1), modify the resistance of the load rheostat to vary the average current delivered between its minimum value (obtained with the rheostat at maximum) and a maximum value of 2 A. Then record the average values of the current delivered by the power supply V_E, I_E, the output current, I_S, and the output voltage, V_S.

2. Deduce the plot of the characteristics $V_S(I_S)$ parameterized by α. Also plot the power transfer characteristics $P_S(P_E)$ and deduce the efficiency.

ETD 29/16/10

Core B66358

- To IEC 63093-6
- For SMPS transformers with optimum weight/performance ratio at small volume
- Delivery mode: single units



FEK0765-U

Magnetic characteristics (per set)

$\Sigma l/A$ = 0.93 mm^{-1}
l_e = 70.4 mm
A_e = 76.0 mm^2
A_{min} = 71.0 mm^2
V_e = 5350 mm^3

Approx. weight 28 g/set

Ungapped

Material	A_L value nH	μ_e	P_V W/set	Ordering code
N27	2000 +30/–20%	1470	< 1.04 (200 mT, 25 kHz, 100 °C)	B66358G0000X127
N87	2200 +30/–20%	1610	< 2.80 (200 mT, 100 kHz, 100 °C)	B66358G0000X187
N97	2250 +30/–20%	1670	< 2.40 (200 mT, 100 kHz, 100 °C)	B66358G0000X197

Gapped (A_L values/air gaps examples)

Material	g mm	A_L value approx. nH	μ_e	Ordering code ** = 27 (N27) = 87 (N87)
N27, N87	0.10 ±0.02	621	457	B66358G0100X1**
	0.20 ±0.02	383	281	B66358G0200X1**
	0.50 ±0.05	201	148	B66358G0500X1**
	1.00 ±0.05	124	91	B66358G1000X1**

The A_L value in the table applies to a core set comprising one ungapped core (dimension g = 0 mm) and one gapped core (dimension g > 0 mm).

Other A_L values/air gaps and materials available on request — see Processing remarks on page 6.

Calculation factors (for formulas, see *"E cores: general information"*)

Material	Relationship between air gap – A_L value		Calculation of saturation current			
	K1 (25 °C)	K2 (25 °C)	K3 (25 °C)	K4 (25 °C)	K3 (100 °C)	K4 (100 °C)
N27	124	–0.7	195	–0.847	181	–0.865
N87	124	–0.7	192	–0.796	176	–0.873

Validity range: K1, K2: 0.10 mm < s < 2.00 mm
K3, K4: 70 nH < A_L < 680 nH